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Nib InternationalBank

NIB CONTROL360

Internal Control Findings Management System (ICFMS)

MASTER IT DEVELOPMENT & IMPLEMENTATION DOCUMENT

NIB International Bank S.C., Ethiopia

Version 1.0 | Functional, Technical, Data, Workflow and Implementation Blueprint

1. Executive Summary

NIB Control360 is an enterprise Internal Control Findings Management System intended to replace the current Excel-based monthly findings reporting, consolidation, review, authorization, rectification and performance-monitoring process with a centralized, role-based and auditable web application.

The system covers the entire bank: 13 districts and more than 410 branches. Districts and branches must be configuration-driven so authorized administrators can add, edit, deactivate or remove them without changing application code.

Business objectives

- Centralize Internal Control and Internal Audit findings.
- Replace manual Excel consolidation and formula adjustment with controlled workflows.
- Provide role-specific dashboards from branch through district and Head Office.
- Track case count and financial amount, including partial rectification.
- Carry unresolved cases between reporting periods without double-counting.
- Use a configurable performance engine; currently Other Case is the scoring category.
- Provide audit trail, threaded comments, optional attachments and notifications.
- Provide controlled monthly reporting periods and management reports.

2. Scope

Area	Requirement
Organization	Bank → District → Branch hierarchy; dynamic administration.
Users/Roles	Administrator, HO Internal Controller, District Internal Controller, District Director, Branch Internal Controller, Branch Manager, Executive Read-only.
Findings	Source, month, district, branch, date, operation area, irregularity, classified case, risk, amount, currency, case count, description, recommendation and optional evidence.
Workflow	Branch Controller → District Controller → HO Controller → Branch Manager → Rectification → Authorization.
Rectification	Weekly progress, partial case/amount rectification and remaining balance.
Transfers	Continuation of unresolved cases with history and no duplicate performance counting.
Performance	Configurable scoring; current basis is Other Case; all categories remain visible.
Dashboards	Role/scope-based KPIs, trends, rankings, risks, categories and work queues.
Administration	Users, roles, districts, branches, sources, categories, scoring rules, periods and settings.
Auditability	Approvals, returns, rejections, rectifications, transfers, comments, attachments and audit logs.

3. Current As-Is Process

1) Branch Internal Controller prepares the monthly findings report. 2) District Internal Controllers correct structure/content and consolidate branches. 3) District sends the consolidated report to Head Office Internal Controllers. 4) Head Office reviews and adjusts formulas/structure. 5) Results return through the district. 6) District communicates findings to Branch Managers. 7) Branch Managers take corrective action and report progress weekly. 8) Unresolved findings continue into later reporting cycles.

Main problems: duplicate/manual entry, formula risk, weak traceability, difficult partial rectification, transfer ambiguity, slow consolidation and limited management visibility.

4. Target To-Be Workflow

Stage	System behavior
Draft	Branch Controller enters finding and may save without submission.
Submit	Finding enters District Controller queue and notification is created.
District review	Approve, Reject, Return for Correction, Comment.
District approval	Moves to Head Office Controller for second approval/review.
Rejection	Moves to rejected queue with reason and audit history.
Return	Routes to Branch Controller for correction; history is retained.
HO approval	Routes to Branch Manager for corrective action.
Rectification	Manager submits full or partial corrective progress, amount and cases.
Authorization	District Controller authorizes where policy requires; unresolved balance remains.
Transfer	Unresolved cases/amount are carried to a later period with transfer history.
Close	Case closes only when eligible remaining cases/amount are fully resolved or authorized closure applies.

5. Roles and Scope

Role	Scope	Responsibilities
Administrator	Bank-wide	Users, roles, districts, branches, sources/categories, scoring rules and settings. Only Admin changes scoring rules.
HO Internal Controller	Bank-wide	Second approval/review, Internal Audit entry/import, bank-wide/district reporting, comments and oversight.
District Internal Controller	Own district	Review/approve/reject/return, consolidation, branch performance and reporting-period control.
District Director	Own district	Oversight, district/branch performance, rankings, trends, comments and reports.

Branch Internal Controller	One branch	Register findings, draft/submit, monitor status and respond to corrections.
Branch Manager	One branch	Receive approved findings and submit corrective/weekly progress.
Executive Read-only	Configured read scope	Read-only management dashboards and reports.

Constraints: each branch has one Branch Manager and one Branch Internal Controller. District and Head Office may have multiple Internal Controllers. A user is assigned to only one branch or district/organizational unit at a time.

6. Finding Data Requirements

Field	Req.	Notes
Finding ID/Reference	Yes	Unique identifier.
Finding Source	Yes	Internal Control, Internal Audit and future configurable sources.
Reporting Month/Period	Yes	Controlled reporting period.
District / Branch	Yes	Validated against organizational scope.
Finding Date	Yes	Date identified.
Operation Area	Yes	Process/area involved.
Type of Irregularity	Yes	Detailed issue type.
Classified Case	Yes	ATM mismatch, ATM long outstanding, IT, dormant, zero balance, CK book, Other Case.
Amount / Currency	Conditional	Financial details.
Number of Cases	Yes	Individual cases represented.
Risk Level	Yes	Low/Medium/High/Critical.
Description	Yes	Finding narrative.
Recommendation	Optional	Corrective recommendation.
Evidence	Optional	Approval must not be blocked solely by missing evidence where authorized physical verification is sufficient.
Status / Audit metadata	Yes	Workflow state, creator, approver, dates.

7. Partial Rectification Rules

A finding may represent multiple similar cases with a combined amount. Rectification must support both case count and amount.

Example	Original	After partial rectification
Cases	3	2 remaining
Amount	ETB 45,000	ETB 35,000 remaining
Rectified cases	0	1
Rectified amount	ETB 0	ETB 10,000

Example: three similar cases of ETB 15,000 + ETB 10,000 + ETB 20,000 = ETB 45,000. If only the ETB 10,000 case is corrected, record 1 rectified case/ETB 10,000 and 2 outstanding cases/ETB 35,000. The remaining balance may transfer and later be rectified.

- Rectified cases cannot exceed eligible cases.
- Rectified amount cannot exceed eligible amount.
- Every rectification is a transaction linked to user, date and period.
- Original finding values must remain auditable.
- Outstanding = eligible original/current quantity minus cumulative authorized rectification.

8. Transferred Case Engine

A transferred case is a continuation, not a new finding. Preserve the original finding ID and transfer chain.

Requirement	Rule
Transfer identity	Keep original finding and full transfer history.
From/To	Store source and destination reporting periods.
Balance	Transfer only unresolved cases and amount.
Performance	Do not double-count the same continuing case in old and new periods.
Case age	Track from original finding date.
Timeline	Show transfers and rectification events.

9. Performance / Scoring

Current business rule: districts are evaluated mainly using Other Case. All finding categories must still be shown for general information, including total, rectified and outstanding values.

Primary current formula: Performance % = Rectified eligible Other Cases ÷ Total eligible Other Cases × 100.

The exact denominator treatment for transferred cases, reporting-period eligibility and approval status must be stored in a versioned Scoring Rule. Do not hard-code these policy decisions.

- Non-scored categories remain visible.
- Admin alone can add/remove/activate scoring categories.
- Scoring rules require version, effective date, active state, included categories/sources and calculation basis.
- District/HO score adjustments, if policy permits, require reason and audit trail.

10. Dashboard Requirements

Role	Required information
Branch Controller / Manager	Selected month; category totals; total/rectified/outstanding; Other Case summary; performance; monthly trend; risk distribution; recent activity; relevant work queues.
District Controller / Director	District aggregate; branch-by-branch performance; top performers; monthly trend; risks; category totals; outstanding; rectification progress; recent activity.
HO Controller	Bank and district aggregates; district/branch ranking; top performers; monthly trend; risks; source comparison; outstanding; rectification; period status.
Executive	Read-only concise KPIs, trends, rankings and exceptions.

Common filters: Reporting Period/Month, District, Branch, Finding Source, Classified Case, Risk, Operation Area and Status. Filters must never bypass organizational scope.

- Monthly Performance Trend
- Risk Distribution
- Finding by Category
- Branch Performance by Month
- District Performance
- Top Performers
- Recent Activity
- KPI cards and exception/work-queue indicators

11. Workflow States

State	Meaning	Typical next state
DRAFT	Not submitted	SUBMITTED
SUBMITTED	Submitted by branch	DISTRICT_REVIEW
DISTRICT_REVIEW	District action pending	APPROVED / REJECTED / RETURNED
DISTRICT_APPROVED	Passed district	HO_REVIEW
HO_REVIEW	Second approval	HO_APPROVED / RETURNED / REJECTED
HO_APPROVED	Approved for branch	SENT_TO_BRANCH_MANAGER
SENT_TO_BRANCH_MANAGER	Awaiting action	PARTIALLY_RECTIFIED / RECTIFIED / TRANSFERRED
PARTIALLY_RECTIFIED	Some resolved	PARTIALLY_RECTIFIED / RECTIFIED / TRANSFERRED
TRANSFERRED	Balance moved	SENT_TO_BRANCH_MANAGER / RECTIFIED
RECTIFIED	All eligible resolved	CLOSED
CLOSED	Terminal	-

12. Comments, Attachments and Notifications

- Threaded comments and replies on findings/workflow actions.
- Users may add attachments to comments where permitted.
- Finding evidence is optional.
- Notifications for submit, approve, reject, return, assignment, rectification, transfer and period events.
- Outlook/email integration should be implemented through a configurable production notification service.

13. Reporting Period Control

- Monthly periods with Open/Locked status.
- District and Head Office Controllers can control periods within authorized scope.
- Locked periods prevent unauthorized reportable changes.
- Lock/unlock requires audit trail and reason.
- Exceptional corrections must be explicit and authorized.

14. Core Data Model

Entity	Key attributes
User	id, name, username, role, status, organizationUnitId,

	districtId, branchId, createdAt, lastLogin
Role	id, code, name, permissions, active
District	id, code, name, status
Branch	id, code, name, districtId, status, managerUserId, controllerUserId
ReportingPeriod	id, year, month, dates, status, lockedBy, lockedAt
Finding	id, reference, source, period, district, branch, date, operation area, irregularity, classified case, amount, currency, cases, risk, description, recommendation, status
Rectification	id, findingId, periodId, rectifiedCases, rectifiedAmount, evidence, comment, submittedBy, authorizedBy, dates
FindingTransfer	id, findingId, fromPeriodId, toPeriodId, casesTransferred, amountTransferred, reason, createdBy, createdAt
Comment	id, findingId, parentCommentId, authorId, text, attachments, createdAt
Evidence	id, finding/comment, file metadata, storage key, uploader, timestamp
Notification	id, recipientId, type, title, message, entityId, readAt, createdAt
AuditLog	id, userId, action, entityType, entityId, oldValue, newValue, timestamp, reason
ScoringRule	id, version, name, active, effectiveFrom, categories, sources, basis, formulaType
ScoringAdjustment	id, target, period, value, reason, adjustedBy, createdAt

15. Recommended Technical Architecture

Recommended production flow: Web UI → API/Application Services → Authentication/Authorization → Relational Database → File/Object Storage → Notification Service → Reporting/Export Service.

Frontend

- React + TypeScript modular architecture.
- Reusable NIB design system.
- Role-aware navigation and route guards.
- UI scope is not a security boundary; server/API must enforce access.

Backend

- Centralized authorization and validation.
- Transactional approval/rectification/transfer operations.
- Idempotency for sensitive actions where appropriate.

Database

- Relational database recommended.
- Indexes for district, branch, period, status, source and category.
- Constraints for one active manager/controller per branch.
- Immutable or protected audit records.

Storage

Use controlled object storage for evidence/attachments and store metadata in the database.

16. Security Requirements

- Production-grade authentication/SSO.
- Server-side authorization by role and organization scope.
- Branch users only see their branch; district users only their district; HO users see bank-wide data as permitted.
- Read-only users cannot modify.
- Audit sensitive actions.
- Never store passwords in plain text.
- Secure attachment validation and access controls.
- HTTPS, session expiry/revocation, secure secrets and logging.

17. Non-Functional Requirements

Area	Requirement
Performance	Optimized for bank-scale historical monthly data and dashboards.
Scalability	No hard-coded 13/410 organization limits.
Availability	Production target and DR to be defined by IT.
Auditability	All critical workflow/configuration actions traceable.
Usability	Responsive, accessible, professional banking UI.
Maintainability	Modular code, API documentation, automated tests and CI/CD.
Data integrity	Transactions prevent invalid partial rectification and duplicate transfers.

18. Reports

- Monthly findings
- Branch performance
- District performance
- Bank-wide performance
- Internal Control vs Internal Audit
- Category summary
- Other Case performance
- Outstanding/unrectified
- Rectification progress
- Transferred/continuing cases
- Risk report
- Top performer/ranking
- Reporting-period status
- Audit trail
- Excel/CSV/PDF/print exports subject to IT standards

19. Administration

Module	Required capability
Users	Create/edit/deactivate/reactivate; role and organizational assignment; active/inactive view.
Districts	Add/edit/deactivate/reactivate.
Branches	Add/edit/deactivate/reactivate; assign one manager and one controller.
Sources	Manage Internal Control/Internal Audit and future sources.
Categories	Manage classified cases.
Scoring Rules	Admin-only configuration/versioning/activation.
Scoring Adjustments	Authorized adjustments with reason/audit.
Reporting Periods	Create, open/lock/unlock within scope.
Settings	Currencies, risk levels, notification settings and other configuration.

20. Testing & Acceptance

Test	Acceptance criterion
Role access	Correct modules/data for every role.
Organization scope	No cross-branch/cross-district unauthorized access.
Finding creation	Mandatory validation; draft and submit work.
Workflow	Branch → District → HO → Manager works with queues.
Return/Reject	Correct routing, reason and history.
Partial rectification	3 cases/45,000 can become 1/10,000 rectified and 2/35,000 outstanding.
Transfer	Only unresolved balance transfers; no duplicate performance counting.
Scoring	Other Case calculation reconciles.
Admin scoring	Only Admin changes scoring configuration.
Period lock	Unauthorized changes blocked.
Audit	Critical actions logged.
Comments/attachments	Threading and optional attachments work securely.
Reports	Totals reconcile to source records.
Build/deployment	Automated build/tests pass.

21. Implementation Roadmap

Phase	Deliverables
1	Architecture, authentication, organization, users/roles, navigation, base UI.
2	Role dashboards, filters, KPIs, trends and aggregation.
3	Finding registration, validation, draft/submit.
4	District review/approval/rejection/return, comments and notifications.
5	HO second approval and Branch Manager routing.
6	Rectification, weekly progress, evidence and partial cases/amounts.
7	Transfer engine, aging, continuing findings and reconciliation.
8	Performance/scoring, configurable rules and

	adjustments.
9	Reports, exports and analytics.
10	Administration, period locks, audit and production hardening.
11	Security, performance, UAT, migration and rollout.

22. Excel Migration

- Create standardized import template aligned to Finding model.
- Validate district/branch codes, periods, categories, currencies and numbers.
- Duplicate detection using approved business keys.
- HO Internal Controllers can import/enter Internal Audit findings.
- Preserve historical source and identifiers where available.
- Reconcile imported totals to approved Excel reports before cutover.
- Document any transformation; never silently change historical performance.

23. Governance

- Business owner: Internal Control Division/designated process owner.
- Technical owner: NIB IT/Digital.
- Security: NIB Information Security.
- Infrastructure/DevOps: NIB IT infrastructure.
- UAT: HO, selected District Controllers, Branch Controllers and Branch Managers.
- Scoring-rule changes require business approval and versioning.
- Production changes follow NIB change management.

24. Final Decisions to Confirm Before Production

- Exact denominator and treatment of transferred cases in performance.
- Whether Internal Audit uses the same workflow or controlled HO entry path.
- Attachment size/types and retention.
- Outlook/email integration architecture.
- Production SSO/identity provider.
- Final report/export standards.
- Data retention/archiving.
- Hosting and disaster recovery targets.
- Exact score-adjustment policy.
- Which partial rectification actions require District authorization.

25. Reference Values

Reference	Values
Roles	Administrator; HO Internal Controller; District Internal Controller; District Director; Branch Internal Controller; Branch Manager; Executive Read-only
Classified Cases	ATM Mismatch; ATM Long Outstanding; IT Case; Dormant Account; Zero Balance; CK Book; Other Case
Risk	Low; Medium; High; Critical

Currency	ETB; USD; EUR; GBP initially; configurable
User Status	Active; Inactive
Period	Open; Locked

26. Production Definition of Done

- All approved workflows work end-to-end.
- Server-side role/scope enforcement.
- Business calculations reconcile with approved Excel/manual results.
- Partial rectification and transfer reconcile by cases and amounts.
- Scoring is versioned/configurable.
- Period locking and audit are operational.
- Security and vulnerability testing complete.
- Performance testing completed with realistic bank-scale data.
- UAT sign-off obtained.
- Backup/recovery/monitoring operational.
- Deployment and rollback plans approved.

END OF NIB CONTROL360 MASTER IT DEVELOPMENT DOCUMENT